

STATE OF THE DATA JOB MARKET

2024 INTELLIGENCE REPORT

Deciphering the signal in 479,000+ global opportunities to decode salary, skill, and hiring benchmarks.

Job Count

479K

Skill per job

4.8

Median Yearly Salary

\$113K

EXECUTIVE SUMMARY: THE 2024 MARKET AT A GLANCE

MARKET DEPTH

478,895

Total Job Postings
Analyzed

COMPENSATION BASELINE

\$113,000

Global Median Salary

TOP SKILL

Python

Required in 51% of
all roles

DOMINANT ROLE

**Data
Engineer**

129,000+ opportunities

The era of the generalist is fading. 2024 data shows a robust market favoring specialized “builders” (Engineers) over pure analysts. While the six-figure salary remains the standard for established roles, access to these roles now requires a specific “tech stack” of approximately 5 distinct skills.

THE DATA FOUNDATION: METHODOLOGY & SCOPE



Raw Data
(CSV/Excel)



Power Query ETL
(Cleaning/Normalizing)



Star Schema Modeling



Final Intelligence

VOLUME

~479k unique records from 2024.

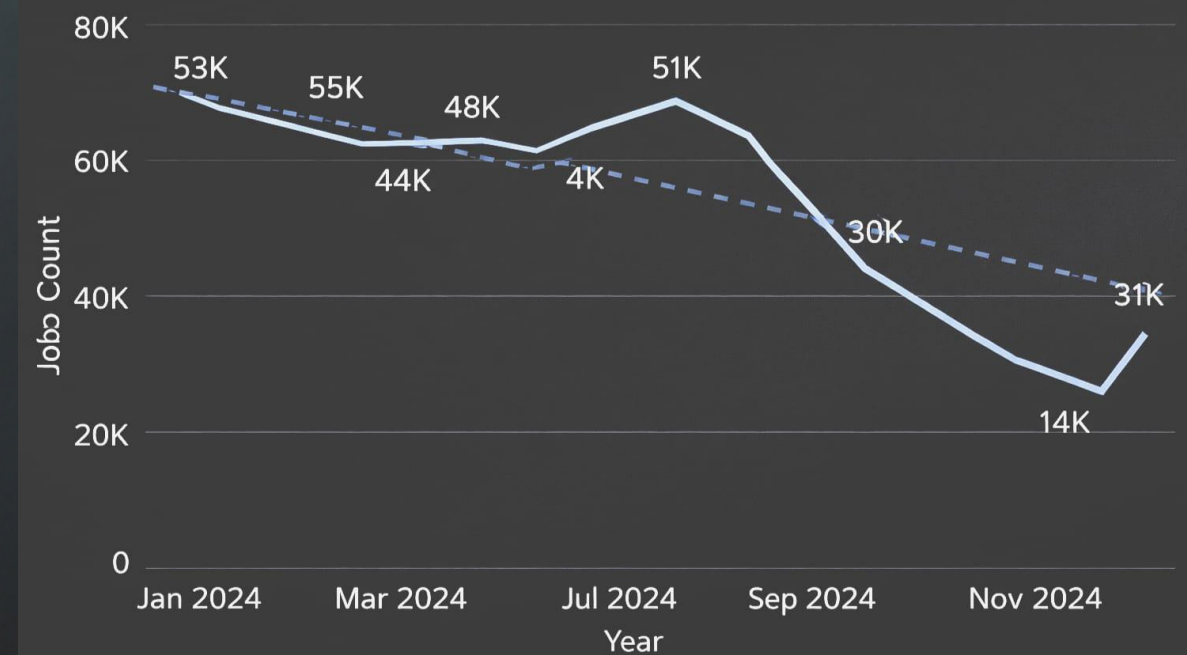
GRANULARITY

17 distinct attributes per job including Salary, Hourly Rate, Skill Requirements, Geography, and Work Mode.

THE GOAL

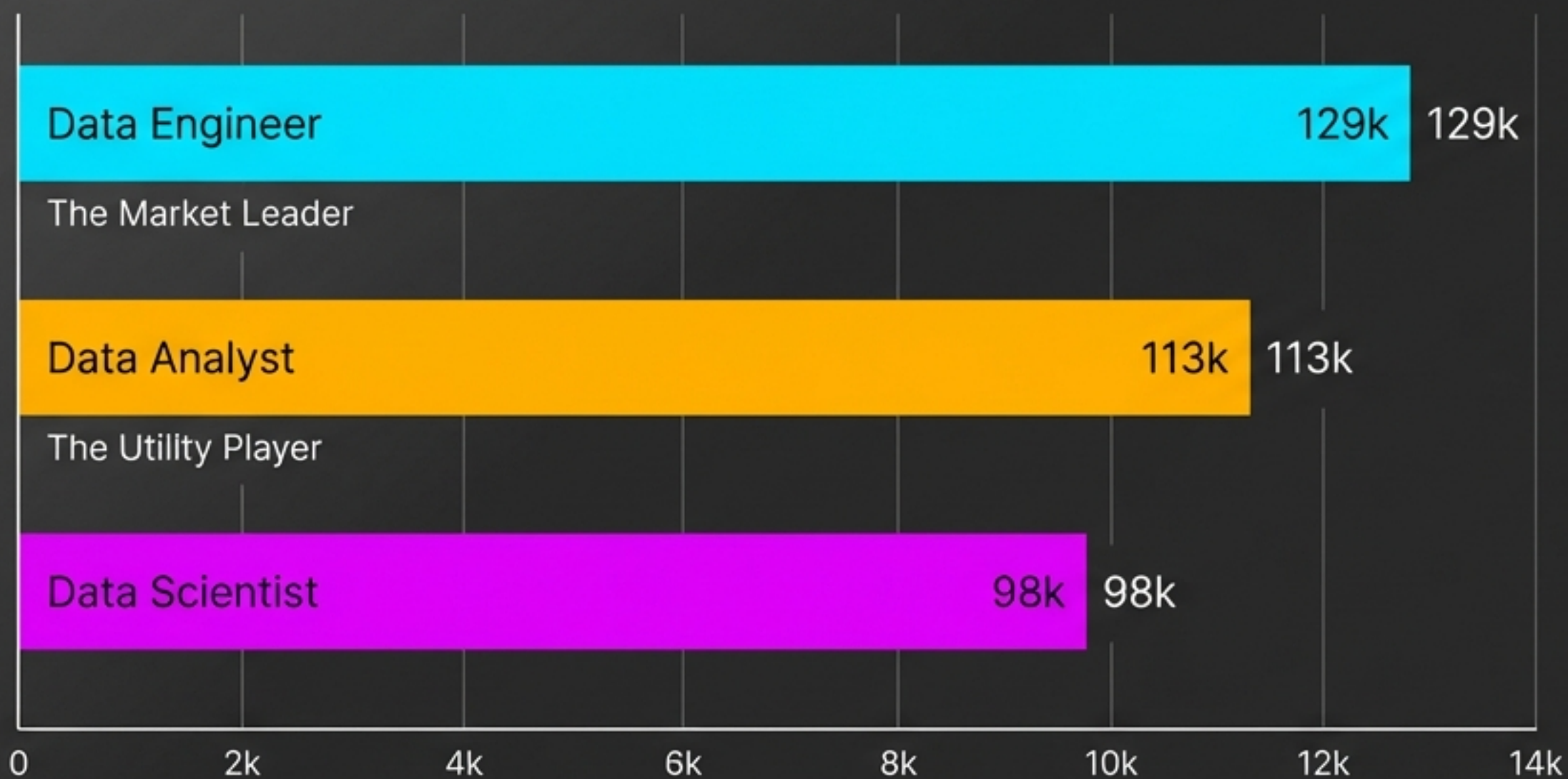
Moving from fragmented noise to consolidated intelligence.

Jobs Over Time



VOLUME VS. HYPE: THE “INFRASTRUCTURE FIRST” REALITY

Job Postings by Role



Insight:

Everyone wants to be a Data Scientist, but the market needs Builders.

Despite the hype surrounding **Data Science**, the market demand for Data Engineers is ~30% higher.

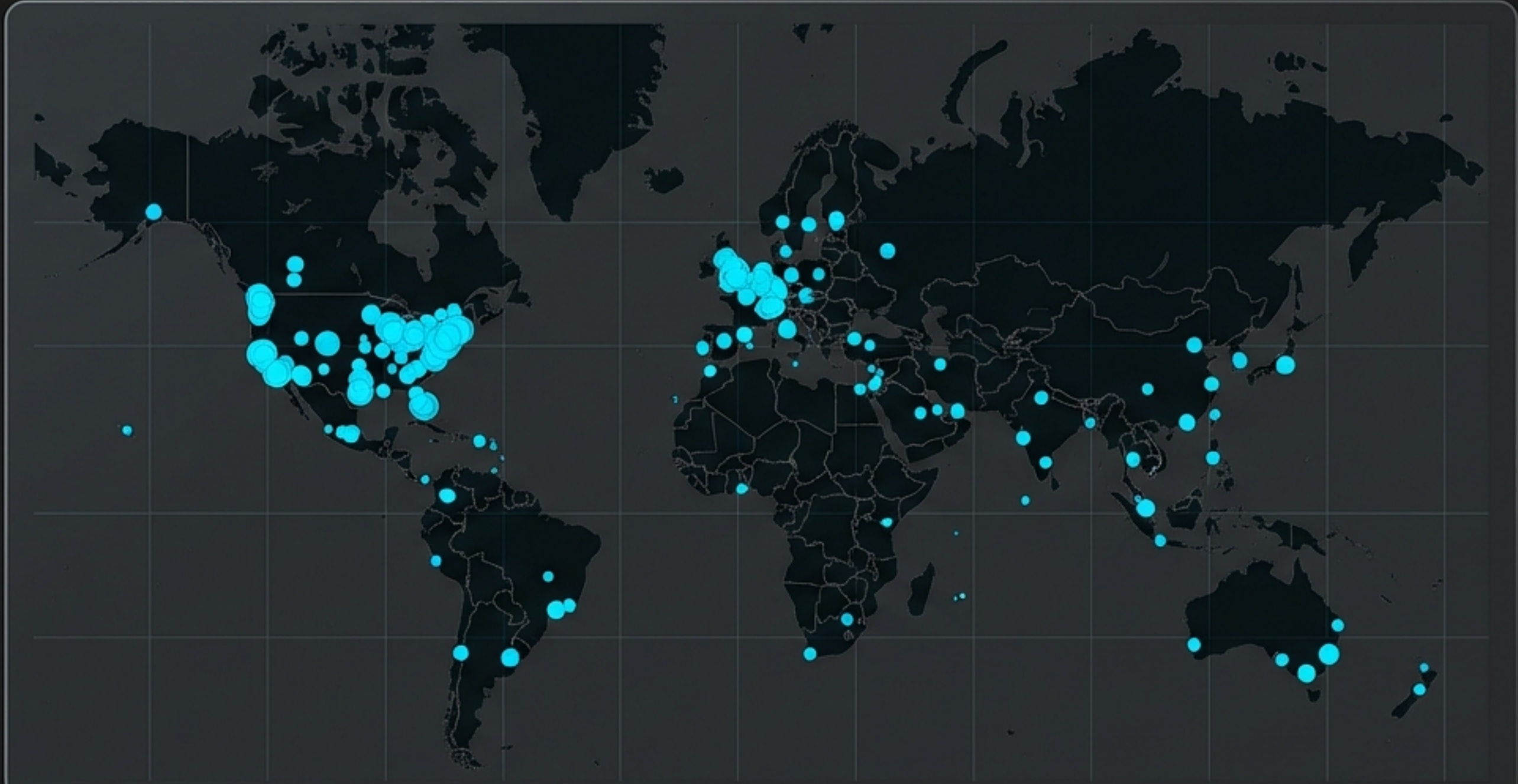
Companies cannot analyze data they cannot move or store. Infrastructure precedes analysis.

THE GEOGRAPHY OF OPPORTUNITY & THE REMOTE PARADOX

THE REMOTE PARADOX

True Remote: ~10%

Hybrid / On-Site: 90%



Key Takeaway: While the work is digital, the career is physical. Proximity to major economic hubs in North America and Europe remains the single strongest lever for maximizing opportunity volume.

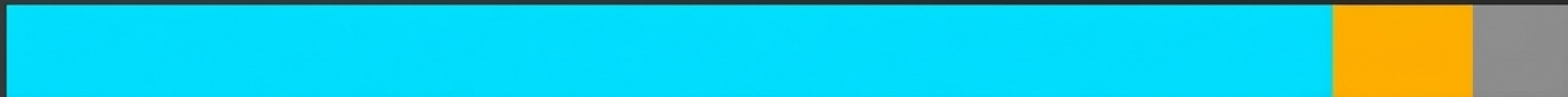
GLOBAL COMPENSATION BASELINES

\$113,000

Median Yearly Salary (USD)

\$48.00

Median Hourly Rate (USD)

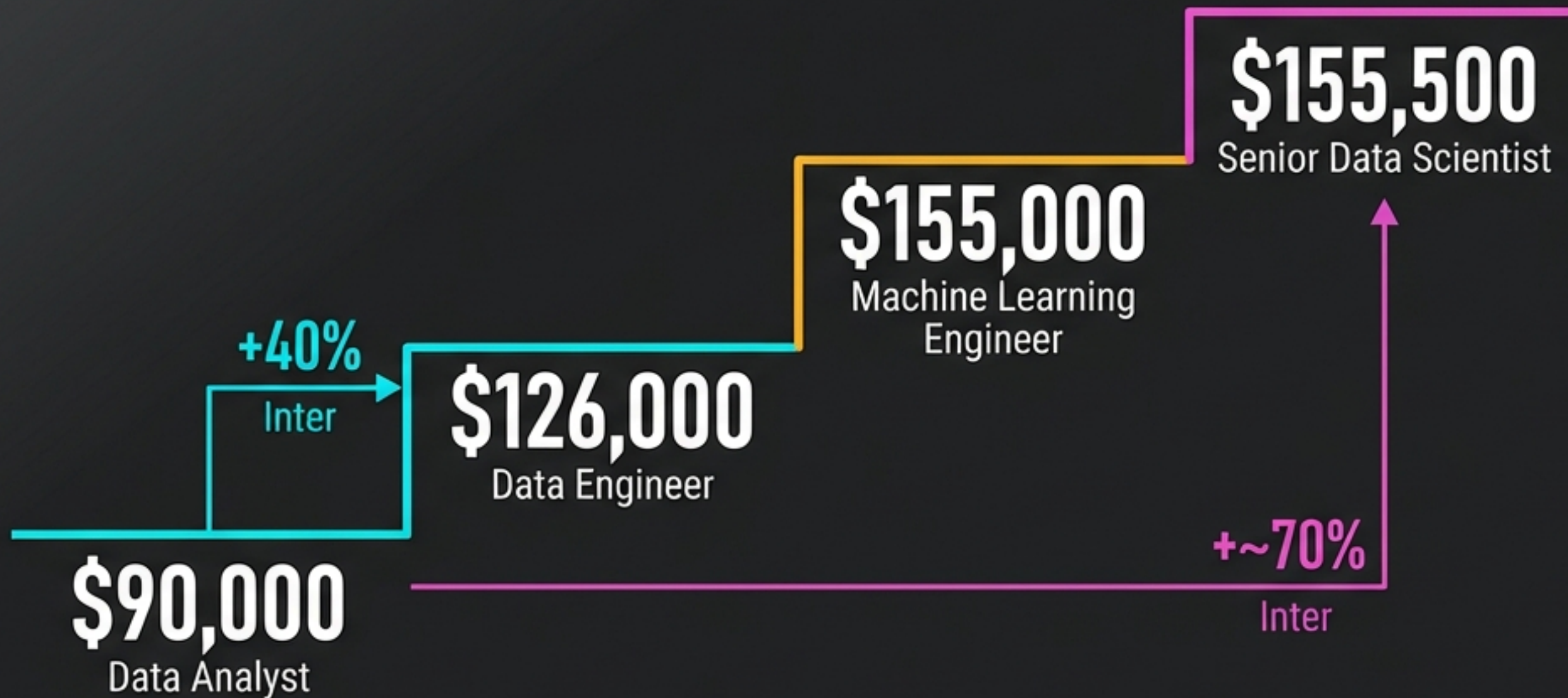


Full-Time: 88%

Contract: 7% Other: 5%

The six-figure salary is not an outlier; it is the median standard for the global data sector in 2024.

THE 'SPECIALIST PREMIUM': VALUING SCARCITY

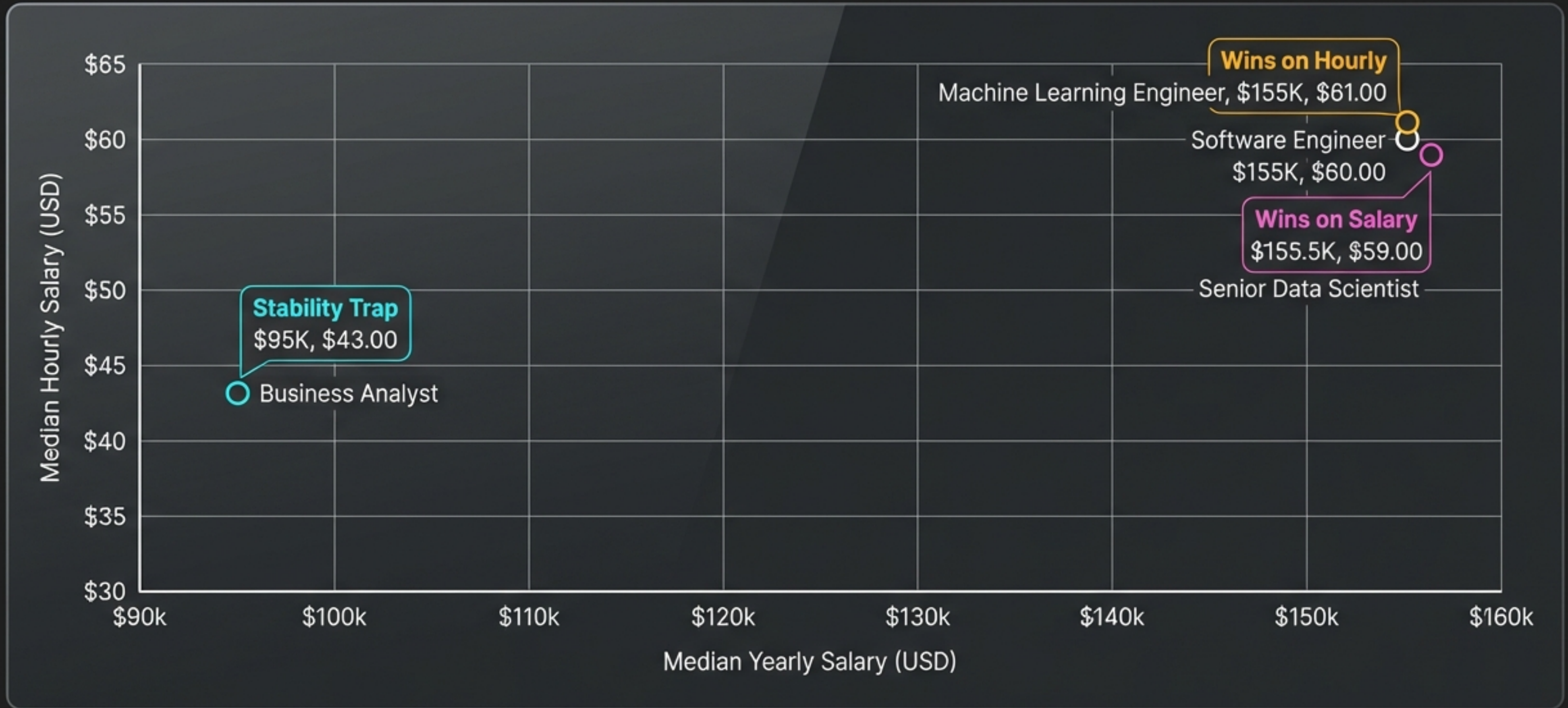


INSIGHT

Moving from 'Analyst' to 'Engineer' yields a massive ROI.

The market pays a premium for those who can build systems (Engineers) and predict outcomes (Scientists/MLEs).

THE HOURLY ANOMALY: STABILITY VS. CASH FLOW



For short-term consulting and contract work, engineering skills (MLE/Software Eng) command the highest immediate market value.

THE 'BIG TWO': MANDATORY TECHNICAL LITERACY

PYTHON

51%

Appears in ~244k jobs

SQL

50%

Appears in ~240k jobs

AWS (21%)

Azure (19%)

Tableau (15%)

Python and SQL are no longer 'nice to haves'. Appearing in 1 out of every 2 job postings globally, they are the foundational language of the industry.

THE '4.8 SKILLS' RULE: THE CLUSTER EFFECT

4.8

Average Skills per Job

The Competitive Stack =



Python (Scripting)

+



SQL (Querying)



AWS (Cloud - **21% of postings**)



Tableau/Power BI (Viz)

+



Domain Skill

Single-skill candidates are at a statistical disadvantage. The market buys 'stacks', not just languages.

PERSONA SPOTLIGHT: THE DATA ANALYST

Subhead: The Utility Player

PROFILE

Role: Data Analyst

Volume:
112,866 Postings



Barrier to Entry:
Low/Medium



COMPENSATION

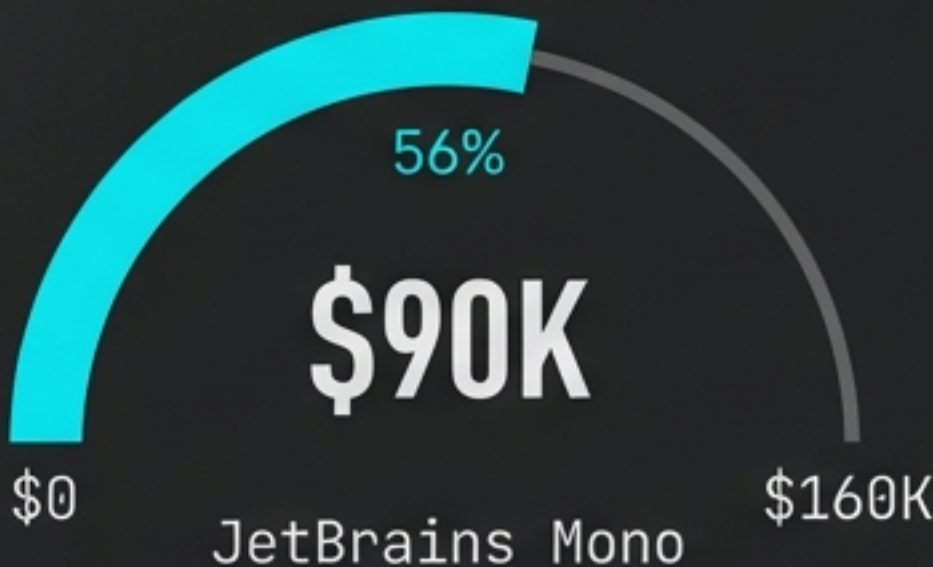
Median Salary

\$90,000

Hourly Rate

\$32.50 Lowest in sector ↘

SALARY TRAJECTORY



STRATEGY

Advice: High volume means easier entry. However, the hourly rate is the lowest in the data field. The path to **\$100k+** lies in rapid upskilling toward Engineering or Senior Analyst roles. ↗ ⚙️

PERSONA SPOTLIGHT: THE DATA ENGINEER

The Market Leader

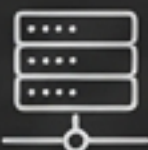
PROFILE

Role: Data Engineer

Volume: **128,994 Postings to Highest Demand**



Role Type:
Infrastructure Architect



COMPENSATION

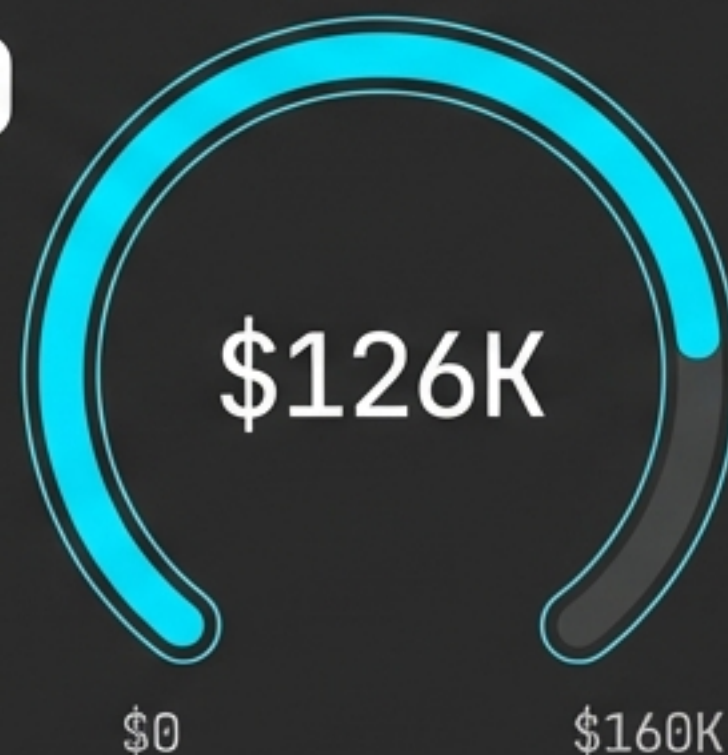
Median Salary

\$126,000

Hourly Rate

\$59.16

SALARY TRAJECTORY



STRATEGY

Advice: The safest strategic bet for long-term career stability. With 30% more open roles than Data Scientists and a \$36k premium over Analysts, it offers the best risk/reward ratio in the 2024 market.



PERSONA SPOTLIGHT: THE MACHINE LEARNING ENGINEER

The High-Value Specialist

PROFILE

Inter

Role:

Machine Learning Engineer

Volume:

~12,900 Postings



Niche/Low Liquidity

Role Type:
Specialist

COMPENSATION

Inter

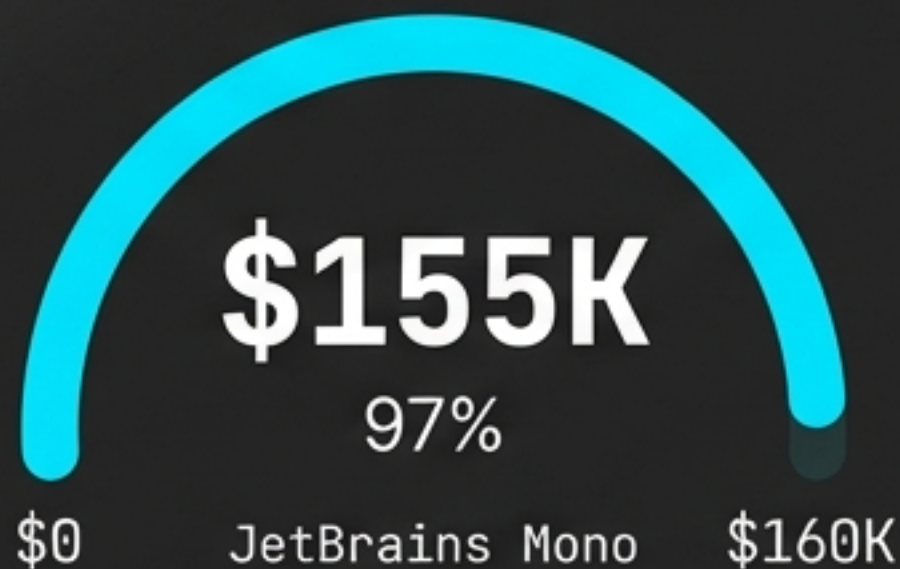
Median Salary

\$155,000

Hourly Rate

\$60.75 ↘ Highest Tier

SALARY TRAJECTORY



STRATEGY

Inter

Advice:

A 'High Value / Low Volume' play.

While lucrative, these roles are scarce (only ~10% of the volume of Data Engineers).

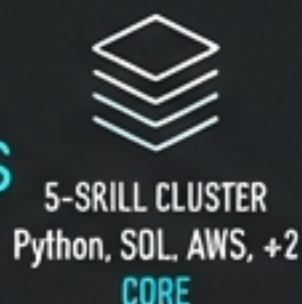
This path requires deep specialization and assumes longer search times.



2024 STRATEGIC PLAYBOOK

👤 FOR CANDIDATES

- **STACK YOUR SKILLS:** Aim for the 'Cluster' of 5. **Python + SQL + AWS** are the non-negotiable core.



- **TARGET THE BUILDERS:** **Data Engineering** offers the highest probability of employment (**129k roles**).



- **RESPECT GEOGRAPHY:** **Hybrid** is the norm. Proximity to hubs increases lead volume.



👥 FOR HIRING MANAGERS

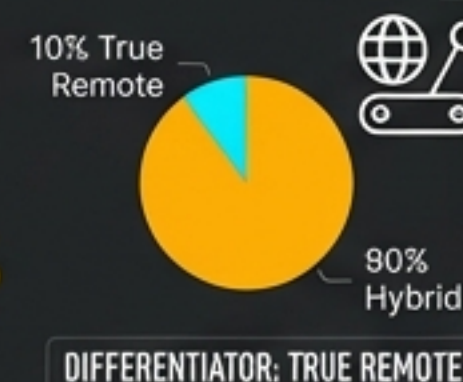
- **BUDGET FOR MEDIAN:** Expect to pay **\$113k** for mid-level talent.



- **THE MLE TAX:** Budget premiums for Machine Learning Engineers; their hourly contract rates top the market.



- **REMOTE IS RARE:** Use **"True Remote"** as a differentiation lever to attract talent in a **90% hybrid market**.



INTELLIGENCE OVER INTUITION

In a market of nearly half a million opportunities, success **isn't about luck—it's about alignment with demand**. The 2024 data is clear: **value has shifted from generalist analysis to specialized infrastructure and engineering**.

TECHNICAL CREDITS

ANALYSIS ENGINE:
Microsoft Power BI

METHODOLOGY:
Star Schema Modeling, DAX
Measures, ETL Transformation

DATA SOURCE:
478,895 Global Job Postings
(2024)